

Multilingual IVR — Architecture

The merged system, and how it compares with the two sources it was built from.

Target multilingual_ivr · **Foundation** voicebot-main (byte-identical copy, MD5-verified; nothing removed)

Behaviour merged from merged_code 1 · **Date** 2026-10-01

Verified state 3,118 tests pass · 0 failures · 63 errors, every one No module named 'nemo' (ASR library deliberately not installed)

1. What this system is

One NVIDIA L4-class GPU runs **three ASR checkpoints and three TTS checkpoints resident at once** — Bengali, Hindi, English — with the language identified from **audio** before recognition starts. Exactly one step in a turn is a language model, and it only decides *what the caller wants*. Every price, date, doctor and confirmation number comes from the clinic database, substituted into fixed wording.

The merge widened the **deterministic guard chain** (Stage 5) from 3 guards to 11. Every new behaviour therefore reaches the caller **before the model runs**, which is why `intent_schema.py` and the LLM prompt could stay byte-identical and the model's accuracy on the existing 14 intents cannot have regressed.

Headline numbers

	merged_code 1	voicebot-main	multilingual_ivr
Languages spoken	1	3	3
agent/ modules	39	94	104
main.py lines	7,985	3,641	3,973
clinic-api routes	28	54	58
Intents (model-visible)	28	14	14
Pre-model guards	5	3	11
Fixed phrases per language	inline	32	40
Test files	75	108	120
Tests passing	–	2,418	3,118
Tests failing	–	17	0
Errors	–	259	63 (all nemo)

Why a function-name comparison is misleading. An inventory said 910 of merged_code 1's 1,192 names were absent from the merge. Acting on that would have duplicated working code: ~600 of them are features **voicebot-main already implements under different names** (`slot_parse` → `intent_schema` , `outcomes` → `outcome_metrics` , and 19 more), and 53 are automation that was deliberately not ported. Every candidate was checked against the running code, not the file list.

2. One caller turn, end to end — multilingual_ivr

```
CALLER (phone / browser)
|
+-----+
| :8100 main.py (WebM) :8101 main_pcm.py (PCM, telephony base)
| 3,973 lines each - byte-identical in every patched region
+-----+
      v
+----- STAGE 1 . AUDIO, LANGUAGE-AGNOSTIC -----+
| admission.py -> caps concurrent calls (~28/process) before anything runs |
| pcm_buffer / vad_stream (Silero) -> "has the caller stopped talking?" |
| audio_quality . echo_cancel . channel_quality . barge_in . playback_gate |
| full_duplex -> half-duplex gate mutes mic while TTS plays |
+-----+
      v
+----- STAGE 2 . LANGUAGE ID -- ON AUDIO, BEFORE ASR -----+
| lid.py SpeechBrain VoxLingua107 . CPU ONLY . per utterance, never latched |
|
| TRAP, measured on the pod: Indian-accented English is labelled Hindi at
| 0.90-0.91 confidence. So the top label is NOT trusted alone --
| lang_select.py: any real mass on a 2nd language (>=0.03) => run TWO ASRs
| and decide by DECODER AGREEMENT, not by LID's own confidence.
+-----+
      v
+----- STAGE 3 . ASR ROUTER -----+
| asr_router.py -- {lang: engine}; construction FAILS LOUDLY if any of
| bn/hi/en has no engine (startup error, never mid-call) |
|
| bn --> IndicConformer (AI4Bharat NeMo fork) in-process
| hi --> IndicConformer (same fork, hi checkpoint) in-process
| en --> FastConformer (MAINLINE NVIDIA NeMo) HTTP :8003, own venv
| - forced split: the two NeMo forks cannot co-exist in one
| interpreter (mainline raises KeyError:'dir' on the Indic ckpt).
| A FILE PATH is passed, not bytes -- both share /workspace.
+-----+
      transcript + detected_lang
      v
+----- STAGE 4 . TRANSCRIPT HYGIENE & STICKY REPLY LANGUAGE -----+
| transcript_rules . speech_norm . bn_normalize . confusables . code_switch |
| language_policy -> reply language is STICKY; only an explicit "speak Hindi" |
| request moves it (ambient LID no longer does) |
+-----+
      v
##### STAGE 5 . DETERMINISTIC GUARD CHAIN -- NO MODEL RUNS HERE #####
# Fixed order, first match wins, each returns. Verified identical in both #
# orchestrators. [+] = added by this merge #
# #
# 1 emergency.py ..... possible medical emergency -> handoff #
# 2 abuse.py ..... swearing -> calm boundary (3x => close) #
# 3 transcript_problem ..... jumbled/empty -> reask_policy #
# ----- #
# 4 clinical_safety.py ..[+].... "is my result dangerous" -> OFFER CLINICIAN#
# 5 complaint.py .....[+].... -> verbatim to DB + route to a person #
# 6 doctor_personal_request[+].. "talk to a doctor" -> decline + offer appt #
# 7 human_request.py ..... "talk to a person" -> immediate handoff #
# 8 callback_request.py .[+].... "call me back" -> queue, never promises when#
# 9 compare_flow.py .....[+].... "which is cheaper" -> Decimal arithmetic #
# 10 doctor_list.py .....[+].... "which doctors" -> list / by-department #
# 11 unverifiable_claim.py[+].... claim it cannot check -> decline + offer #
# ----- #
# 12 _update_caller_state -> anger.py [+] #
# sets caller_state="angry" => speech_policy row that ALREADY EXISTED #
# but nothing ever set: slower rate, 1 question/turn, offer_human, #
# escalation_threshold=low, + trilingual single-apology acknowledgement #
#####
# (nothing matched) v
+----- STAGE 6 . INTENT -- CHEAPEST TIER THAT ANSWERS WINS -----+
| TIER 1 fast_path.py + fast_path_cues.py string/entity match, NO model |
| per-language cue tables . COMMIT_FLOOR 0.72 . abstains if unsure |
| | miss |
| TIER 2 semantic_cache.py bge-m3 embeddings via Ollama :11434 |
| caches the INTENT, never the answer => a changed price stays live |
| | miss |
| TIER 3 llm.py -> Qwen2.5-7B Ollama :11434 . strict JSON mode |
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+-----+
| THE ONLY NON-DETERMINISTIC STEP IN THE WHOLE TURN. |
| It picks WHAT the caller wants. It is never shown a price, |
| a date or a confirmation number it could restate. |
| Output parsed by intent_schema.py (pydantic, 14 intents) -- |
| UNTOUCHED by this merge, so its accuracy cannot regress. |
+-----+

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~65-75% of turns never reach Tier 3

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+----- STAGE 7 . GATES BEFORE ANY ACTION OR FACT -----+
| action_gate.py -- ONE place decides: act | ask | read back | repeat | hand |
| confidence_gate . entity_confirmation . identity/security_check (2-factor) |
| speaker_change -> notices a different voice after verification |
+-----+

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+===== STAGE 8 . SOURCE OF TRUTH -- clinic-api :8080 =====+
| FastAPI + SQLite . 58 routes (54 + 4 added) . live lookup, NEVER cached |
| tools_client.py validates every answer against api_models.py (pydantic) |
| money parsed with parse_float=str => "450.00" never becomes "450.0" |
| |
| [+] POST /api/v1/complaints          [+] GET /api/v1/departments |
| [+] GET /api/v1/doctors              [+] GET /api/v1/doctors/by-department |
| [FIX] idempotency.py __signature__ -- the app could not even IMPORT before |
+-----+

```

verified fact

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+----- STAGE 9 . REPLY ASSEMBLY -- TEMPLATE SUBSTITUTION -----+
| reply_templates.py (bn, 560 ln) --dispatch--> reply_templates_i18n (hi/en) |
| phrases.py 40 fixed non-factual lines x bn/hi/en, coverage pinned by tests |
| verbalize() . pronunciation lexicon . spoken_text_lint (artefact RATCHET) |
| persona.py register . no hedging . NO REASSURANCE . sentence-length caps |
| (bn 15 / hi 18 / en 20 words) |
| apology.py AT MOST ONE apology per spoken reply |
+-----+

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+----- STAGE 10 . TTS ROUTER -----+
| tts_router.py -> tts_server.py :8002 -- ONE coqui-tts process, ALL 3 voices |
| AI4Bharat Indic-TTS (FastPitch + HiFi-GAN) ships bn/hi/en in one format, |
| so unlike ASR no second TTS stack is needed for English. |
| clause_split -> short clauses, streamed . prewarmed phrase cache |
+-----+

```

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+----- STAGE 11 . RECORD (counts and ids only, never free text) -----+
| call_record . call_score (anger_turns[+], clinical_questions[+], |
| complaint_filed[+]) . outcome_metrics . latency_metrics . call_summary |
| EXCEPTION, by design: the complaints table holds the caller's VERBATIM |
| words -- one narrow-access store, so logs stay text-free. |
+-----+

```

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CALLER

3. Three-way comparison

	merged_code 1 (Bengali only)	voicebot-main (trilingual base)	multilingual_ivr (= base + merge)
=====			
LANGUAGE			
audio LID	none	SpeechBrain, CPU	same
ASR	1 x Bengali	3 routed (bn/hi/en)	same
TTS	1 x Bengali	3 voices, 1 process	same
lang detection	from TRANSCRIPT	from AUDIO, pre-ASR	same
reply languages	bn only	bn + hi + en	same
STRUCTURE			
agent/ modules	39	94	104 (+10)
main.py	7,985 lines	3,641 lines	3,973 (+332)
logic lives in	main.py + one *_flow.py per story	~94 small modules behind a gate layer	same, + 10 more guards
reply_templates	4,849 ln (bn only)	454 ln + i18n pair	560 ln + i18n
phrase table	inline per function	32 keys x 3 langs	40 keys x 3
INTENT			
intents	28	14 (pydantic Literal)	14 UNCHANGED
resolution	fast_path -> cache -> LLM	3 tiers + abstention + decoder agreement	same
new behaviour reaches caller by	new intents	--	DETERMINISTIC GUARDS, pre-model
CALLER-FACING BEHAVIOUR			
anger	yes anger_flow	NO - policy existed DEAD	YES detector wired
complaints	yes complaint_flow	NO	YES + DB + endpoint
clinical safety	yes clinical_safety	NO	YES + Devanagari
doctor personal	yes	NO	YES
unverifiable	yes 4 categories	NO	YES 2 (other 2 are checkable here)
callback ask	yes (LLM intent)	NO - queue, no detector	YES detector added
doctor list	yes 423 ln extractor	NO intent, NO endpoint	YES 150 ln + 3 eps
compare options	yes compare_flow	partial (pkg-vs-separate)	YES Decimal money
answer ledger	yes	NO	code only, unwired
date_calc	yes code owns dates	NO - model resolves dates	SKIPPED (needs prompt change)
reports + OTP	yes own design	yes own design	base kept
cancel charges	yes charging_for_	yes cancellation_charge()	base kept
correction / silence / symptom / speakability / confidence / match-band /			
turn-log	yes	yes (own implementations)	base kept
		`- 21 modules NOT ported: would duplicate	
SAFETY / TRUTH			
truth boundary	templates only	templates only	same
pre-LLM guards	5	3	11
model-written	smalltalk only	smalltalk, filtered by	same
text	(unfiltered)	persona+speakeable+history	
money arithmetic	Decimal	none performed	Decimal (1st use)
ENGINEERING			
clinic-api routes	28	54	58
test files	75	108	120
tests passing	--	2,418 (17 fail/259 err)	3,118 (0 fail/63 err)
CI / gate	none	gate.sh (COULD NOT RUN)	gate.sh RUNS
ADR / CLAUDE.md	none	yes	yes
deploy	vast.ai	RunPod, single L4	RunPod (vast NOT ported, as asked)

4. Process topology — why three Python environments

Identical in voicebot-main and multilingual_ivr. merged_code 1 had one environment and one language.

```

+-- /workspace/venv -- AI4Bharat NeMo FORK -----+
|
| :8100 main.py      :8101 main_pcm.py      |
|      orchestrator      orchestrator (telephony) |
|
| bn ASR  in-process      hi ASR  in-process      |
|
| :8080 clinic-api  FastAPI + SQLite      |
|      the ONLY source of any price/date/doctor |
+-----+
|
|      | HTTP, WAV path only      |
|      v                          v
+-- /workspace/venv-en-nemo --+  +-- /workspace/tts_venv -----+
| MAINLINE NVIDIA NeMo      |  | coqui-tts      |
| :8003 english_asr_server  |  | :8002 tts_server - bn + hi + en |
|      en ASR FastConformer|  |      ONE process, all 3 voices |
+-----+  +-----+

```

THE ISOLATION IS FORCED, NOT STYLISTIC:

- ```
* IndicConformer (bn/hi) loads ONLY under AI4Bharat's NeMo fork;
mainline raises KeyError: 'dir' on that checkpoint.
* FastConformer (en) needs MAINLINE NeMo -- the two forks cannot both be
importable in one interpreter. Hence English ASR is a 4th process.
* coqui-tts needs huggingface_hub>=0.34; the AI4Bharat fork imports
ModelFilter, REMOVED in 0.24. No single version satisfies both.
=> No version pair satisfies all three. Separate venvs, native processes,
no Docker.
```

## 5. The merge as a delta

```
voicebot-main -----> multilingual_ivr

agent/ 94 modules --[+10 new]-----> 104
 anger, complaint, clinical_safety,
 doctor_personal_request, unverifiable_claim,
 callback_request, callback_config,
 doctor_list, compare_flow, answer_ledger

main.py 3,641 ln --[+332]-----> 3,973
main_pcm.py 3,651 ln --[+332]-----> 3,983
 11 imports . 3 session fields
 2 slot_answer entries . 8 guard blocks
 . 1 helper

phrases.py 32 keys x3 --[+8 keys x3]-----> 40 keys x3
clinic-api 54 routes --[+4]-----> 58
models.py --[+ComplaintRecord]---> new complaints table
idempotency.py BROKEN --[__signature__ fix]--> app imports (62 routes)

tests 108 files --[+12]-----> 120
 2,418 pass -----> 3,118 pass
 17 fail -----> 0 fail
 259 error -----> 63 error (all 'nemo')

intent_schema.py 14 intents --[UNTOUCHED]-----> 14 intents
llm.py prompt --[UNTOUCHED]-----> unchanged
`- so Owen's accuracy on the existing intents cannot have regressed
```

## 6. What the diagrams are really saying

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### The language layer is entirely voicebot-main's, untouched

LID-before-ASR, the two-decoder disambiguation for accented English, three routed engines, one TTS process for all three voices. The merge added nothing here and broke nothing here. merged\_code 1 had none of it — it detected language from the Bengali *transcript*, which stops working the moment audio must be routed to one of three engines, because there is no transcript yet at the point the decision is needed.

### The merge is a widening of Stage 5, not a redesign

Eight new guards sit in the deterministic chain ahead of the model. Two orderings are load-bearing and are asserted in tests rather than left as comments:

- **clinical before complaint** — "I want to file a complaint, is my result dangerous" takes the clinical path; the health question must not go unanswered, and that path reaches a person too.
- **complaint before the request-for-a-person** — a deliberate divergence from merged\_code 1. Exactly one utterance collides: "I want to speak to someone about a complaint". Under the original order it is routed as a bare person-request and **the complaint is never recorded** — the call still looks fine, which is why it was worth changing.

### merged\_code 1's shape did not survive, deliberately

One 7,985-line `main.py` with a `*_flow.py` per story became detectors feeding voicebot-main's existing gate, policy and template machinery. Two concrete consequences:

- **Anger cost ~150 lines instead of 220.** voicebot-main already contained a complete, trilingual `"angry"` speech policy and acknowledgement — and **nothing ever set it**. A grep returned three definition sites and nothing else. The feature existed, in three languages, and was unreachable. Only the detector was missing.
- **The doctor list cost ~150 lines instead of 423.** The original carried its own department-name extractor — plural detection, Bengali case endings, fuzzy scoring with its own floor and margin. All of it was a text matcher in `agent/` guessing at a name it had no list to check itself against. The new endpoint matches the caller's words against the real department table in both directions, so the clinic's own data decides and no extractor is needed.

### Known limits, stated plainly

- **No Hindi or Bengali audio has gone through any of the new paths.** The code paths are tested offline; the voices are not. Every Devanagari phrase list was written for this merge and needs a native speaker's review.
- **Department names are Latin-only** in the schema, so a Bengali or Hindi reply names a department in a script that voice drops. Pre-existing — `reply_templates.py:390` already does this — and the fix is localised department-name columns.
- **answer\_ledger is code only.** Wiring it needs a signature change across 12 call sites and 9 test files, which belongs in its own review.
- **date\_calc was skipped.** It would move calendar arithmetic out of the 7B model, which is a real truth-boundary win, but its design needs a new prompt slot ( `date_expr` ) and the prompt was out of scope.